

ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-CE (revision of PROJECT-ASYNC)
ORIGINATION: Prof. Kidd, Instructor (ECO 0.1, 1/1/2030)
PURPOSE OF CHANGE: Cyclic executive task manager
DOCUMENT NUMBER: ECN0.1
DATE OF PUBLICATION: October 3, 2026

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EXECUTIVE SUMMARY: This ECN implements ECO 0.1 by replacing the inline scheduling in the PROJECT-ASYNC design with a polled, round-robin cyclic executive. The serial dialog, the LED1 blinker and the LED2 blinker are now three task functions listed in a function pointer table, `taskTable[]`. The `loop()` function calls one task per pass and advances to the next table entry, so each task runs once per three passes. The customer requirements, high-level technical requirements, hardware and user-visible behavior are unchanged. The System Design Document is revised to Rev 2, and one educational constraint (EC.6) and one test (T.7) are added.

HIGH-LEVEL DESCRIPTION: This ECN notifies stakeholders that PROJECT-CE dispatches its tasks from a round-robin cyclic executive instead of the inline code used in PROJECT-ASYNC.

REASON FOR CHANGE: ECO 0.1 requires a more reliable design that reduces labor and design costs for future extensions. A new task is now one new function plus one new table entry, and the dispatcher does not change.

CUSTOMER REQUIREMENTS:

The customer requirements, C1 through C6, are identical to those of PROJECT-ASYNC. See [1] REQUIREMENTS section, PART I, Table 1.

HIGH-LEVEL TECHNICAL REQUIREMENTS:

The high-level technical requirements, HL.1 through HL.6, are identical to those of PROJECT-ASYNC. See [1] REQUIREMENTS section, PART II, Table 3. The educational constraints, EC.1 through EC.5, are also identical; see [1] REQUIREMENTS section, PART II, Table 4.

Changed Documents: provost.cmpe311.fall26.project#2 (System Design Document, Rev 2, revising provost.cmpe311.fall26.project#1 [1]). Deleted text is struck out and additions are marked, each with the date.

- **10/3/26 (REQUIREMENTS, Table 2)** EC.6 added: "The Developer must dispatch The System's tasks using a polled, round-robin cyclic executive task manager based on a function pointer table." (ECO 0.1, Educational Requirement 1.0)
- **10/3/26 (TESTING AND VALIDATION REQUIREMENTS, Table 4)** T.7 added: "The code must dispatch all tasks from a function pointer table in round-robin order, and no task may block." Verifies EC.6 and EC.5. T.1 through T.6 are unchanged and re-run on the PROJECT-CE build.
- **10/3/26 (REQUIREMENTS TRACEABILITY, Table 5)** EC.5 row: verified by T.6, **T.7** (T.7 added). EC.6 row added: verified by T.7 (inspection).
- **10/3/26 (SYSTEM OVERVIEW)** ~~The main loop checks for new serial input and updates both LEDs' blink states on every pass;~~ **The main loop is a dispatcher that calls one task per pass (serial**

dialog, LED1, or LED2) from a function pointer table in round-robin order, so serial communication and LED blinking run concurrently without either blocking the other.

- **10/3/26 (SOFTWARE DESIGN)** ~~The program uses the standard Arduino setup()/loop() structure with no blocking calls;~~ **The program uses a cyclic executive: askQuestions(), serviceLED1() and serviceLED2() are listed in taskTable[], and loop() calls taskTable[currentTask]() and then advances currentTask modulo NUM_TASKS**, so serial communication and LED blinking run concurrently, satisfying HL.4 and EC.5. The serial dialog, LED timing and input rules are otherwise unchanged.
- **10/3/26 (DESIGN RATIONALE AND ASSUMPTIONS)** Added: why one task per pass, why timing still holds, the 1 to 32,767 ms interval range, and the 49.7-day run limit.
- **10/3/26 (APPENDIX B, DESIGN CODE)** PROJECT-ASYNC source replaced by the PROJECT-CE source, which adds the TaskFunction typedef, taskTable[], NUM_TASKS, and the three task functions. The blink and serial functions have no logic change.
- **10/3/26 (TITLE AND DATE, REFERENCES AND GLOSSARY, TESTING RESULTS, APPENDIX A)** Document reference changed to provost.cmpe311.fall26.project#2; revision history, references and new terms added; results and evidence re-recorded for the PROJECT-CE build.

Sections not listed above (customer requirements, HL.1 through HL.6, EC.1 through EC.5, bill of materials, hardware design, pin assignments, T.1 through T.6) are unchanged and are cited to [1] in the revised document.

Added Documents: ECN0.1, provost.cmpe311.fall26.project#2.ECN (this note), added 10/3/26.

PROJECT-CE source sketch added to the GitHub repository

<https://github.com/raheemprovost/CMPE311Project2> on 10/3/26.

Removed Documents: None.

Replaced Documents: None. provost.cmpe311.fall26.project#1 [1] remains the PROJECT-ASYNC baseline.

Any Additional Design Considerations: No hardware change: the board, LEDs, 330 Ω resistors and pins 2 and 3 are as in [1]. Each task now runs once per three loop() passes instead of every pass, which adds a delay far below the shortest practical blink interval. Entered intervals must be 1 to 32,767 ms because they are parsed into a 16-bit int, and continuous operation is supported for up to about 49.7 days because the millis() comparison is not rollover-safe.

REFERENCES:

[1] provost.cmpe311.fall26.project#1, CMPE311 Project 1 design document (PROJECT-ASYNC), Revision 1, September 22, 2026.

[2] provost.cmpe311.fall26.project#2, CMPE311 Project 2 design document (PROJECT-CE), Revision 2, October 3, 2026.

[3] PROJECT#2: PROJECT-CE assignment, including Engineering Change Order ECO 0.1 (1/1/2030, requestor Prof. Kidd), CMPE311.